Mining Human–AI Collaboration in Software Engineering Repositories

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***Abstract*—AI coding agents can now write and modify code and open pull requests (PRs) on GitHub like human contributors. However, it is still unclear how their contributions compare to human ones in terms of development outcomes and contribution characteristics. In our study, we use the AIDev-pop subset of the AIDev dataset, which contains 33,596 PRs from five agents, and compare them with 5,565 human PRs from 818 repositories with more than 500 stars. After cleaning the data and separating human PRs from bot and agent PRs, we compare both groups on a matched subset of 7,541 agent PRs from the same repositories and period as the human PRs, using non-parametric tests, effect sizes and a logistic regression. Agent PRs are merged far less often than human PRs (59.5% vs. 85.1% of resolved PRs), and this gap remains after controlling for task type and PR size (odds ratio of 0.25). However, regarding the other outcomes, the merged agent PRs have a similar number of commits and are merged slightly faster, and both groups receive similar review activity. Both groups also have similar size and scope, while agents do somewhat less CI/CD and maintenance work. Finally, we show that aggregate merge rates overstate agent acceptance, largely because of repositories without human PRs.**

***Index Terms*—AI coding agents, pull requests, mining software repositories, empirical study, code review**

I. Introduction

Since the public release of ChatGPT in November 2022, generative AI has reshaped the daily routines of professionals and students alike. While initial adoption was often cautious, it is now evident that AI is cementing its place in almost every modern workflow. This reality is especially pronounced in software engineering, where AI agents can execute a wide variety of tasks, greatly multiplying the output a single developer can achieve. In fact, a 2023 controlled experiment showed that developers using GitHub Copilot completed a programming task 55.8% faster than those without it [1]. Given the rapid advances in the underlying models over the past three years, these productivity gains have likely continued to grow.

We are currently witnessing a shift from simple autocomplete utilities to semi-autonomous "AI teammates" that are capable of tackling complex engineering tasks independently [2]. As agents such as Claude Code, Devin, OpenAI Codex and GitHub Copilot's coding agent begin opening their own pull requests (PRs), pushing commits, and resolving issues directly within the codebase, understanding their operational impact is critical. These AI teammates matter to modern software engineering not only because they accelerate development, but because they fundamentally alter team collaboration, code maintenance, and project lifecycles. As they assume a larger share of repository activity, it is essential to empirically investigate how their contributions compare to those of human developers in terms of size, scope, and overall development outcomes.

II. Research Questions

We address two research questions.

RQ1: *Development outcomes.* How do AI-related contributions differ from human contributions in terms of development outcomes?

RQ2: *Contribution characteristics.* How do AI-related and human contributions differ in terms of size, scope, and type?

III. Methodology

A. Data mining

For the extraction, we use the AIDev-pop subset of AIDev v3 [2]: 33,596 PRs opened by five agents in repositories with more than 100 stars. The AIDev v3 was chosen instead of v4, because it matches the human PR tables provided by the lab instructor, which contains 6,618 PRs created between January and June 2025 in 818 repositories with more than 500 stars. We use the PR, timeline, review, commit and issue tables of both datasets, manually aligned to a common

schema. Replication package: https://github.com/Tristanlach/LOG6307E-Replication-Project-MSR.

B. Data cleaning

We remove 56 human PRs that duplicate an agent PR (same URL), with all their linked rows, as well as exact duplicate rows. We only keep missing values when they are meaningful. For example, an empty `merged_at` time means the PR was not merged. PRs without any file commit details in the matched subset (168 agent and 15 human PRs) are excluded from size, scope and file type metrics. After cleaning the dataset, it contains 33,596 agent PRs and 6,562 PRs from the human dataset.

C. Classification

For agent PRs, we use the AIDev labels, after checking that each agent's traces (signatures or accounts) appear in 80% to 100% of its PRs. Human PRs are not all written by humans, so we separate them from their author and from agent signatures in descriptions and commits. The Human group keeps the 5,565 human-only PRs and human PRs with some agent commits, and excludes 997 bot and agent PRs, even if they were in the human dataset. For reviews, we only count those written by human accounts, since agents review their own PRs without any human intervention. Human reviews are only visible when they leave inline comments, so we apply the same rule to both groups. The two groups also come from different contexts. Human PRs come from more popular repositories and an earlier period, whereas 92% of agent PRs were created between May 2025 and July 2025. In addition, one personal repository holds 27% of all agent PRs. Therefore, we compare human PRs with 7,541 agent PRs from the same repositories and period. On this subset, the agent merge rate between resolved PRs drops from 77% to 60%. Finally, AIDev assigns each PR one Conventional Commit type with an LLM. Most types map directly to the assignment's categories. We group the PRs into these categories: `fix` as bug fixes, `feat` as feature implementation, `docs` as documentation, `refactor`, `style` and `perf` as refactoring, `test` as testing, `ci` and `build` as CI/CD, and `chore`, `revert` and `other` as maintenance.

D. Metrics and statistical analysis

All metrics are computed per PR on the matched subset. For RQ1, we use resolved PRs (merged or closed), since open PRs have no outcome yet. We measure the merge rate, the number of human reviews, and the share of revised PRs (PRs with at least one review request or reopening event). In merged PRs, we measure the time from creation to merging and the number of commits. For resolved PRs related to an issue, we measure whether the issue was closed and the time from its creation to its closing. For RQ2, we use PRs with file-level commit data and measure lines added and deleted, code churn (added + deleted, summed over all commits), the number of distinct files changed, and whether the PR touches documentation, test, or configuration files, identified from file names.

For yes/no metrics such as merging, we use a chi-square test and report Cramér's V as the effect size. Counts and durations are very skewed, so we compare them with a Mann-Whitney U test and report the rank-biserial r. Since we run several tests for each research question, we adjust the p-values with the Holm correction. With more than 13,000 PRs, even very small differences are significant, so we focus on effect sizes, using as thresholds: 0.1 (small), 0.3 (medium) and 0.5 (large).

Agent PRs may be merged less often simply because they contain different tasks or larger changes. To check this, we fit a logistic regression that predicts whether a PR is merged from its group, its task category, and its size (log of lines and files changed). Because PRs from the same repository are not independent, we cluster the standard errors by repository.

IV. Results

A. RQ1: Development Outcomes

To answer RQ1, Tables I and II present the development outcomes based on the 6,947 agent and 5,155 human resolved PRs of the matched subset. The biggest difference between the two groups is in the acceptance rate, as human PRs have a higher merge rate than agent PRs (85.1% vs. 59.5%), a significant difference of 25.6 percentage points ($p_{\text{Holm}} < 0.001$) with a small-to-moderate effect size ($V = 0.28$). Table II shows that after controlling for task category and PR size, agent PRs still have much lower odds of being merged than human PRs, with an odds ratio of 0.25 (95% CI: 0.21–0.31, $p < 0.001$). Furthermore, the merge rate gap shown in Fig. 1 is significant in all seven task categories after the Holm correction, with effect sizes ranging from $V = 0.09$ for documentation (80% vs. 88%) to $V = 0.34$ for bug fixes (56% vs. 88%).

Other development outcomes in Table I show smaller differences between AI and human-made PRs. The revised PR rates are very close (41.3% vs. 42.9%, $p_{\text{Holm}} = 0.17$, $V = 0.02$), while slightly fewer agent PRs receive at least one human review (18.1% vs. 21.1%, $p_{\text{Holm}} < 0.001$, $r = -0.04$). Among merged PRs, both groups have a median of 2 commits before merge ($p_{\text{Holm}} = 0.17$, $r = 0.02$), while agent PRs are merged a little faster (2 vs. 4 hours, $p_{\text{Holm}} < 0.001$, $r = -0.09$). Finally, for resolved PRs linked to an issue, the share of closed issues is similar (83.7% vs. 87.5%, $p_{\text{Holm}} = 0.06$, $V = 0.05$), though linked issues are closed almost twice as fast for agent PRs (218 vs. 402 hours, $p_{\text{Holm}} = 0.001$, $r = -0.12$).

B. RQ2: Contribution characteristics

To answer RQ2, Table III compares the size, scope, and touched file types of the 7,373 agent and 5,550 human PRs of the matched subset that have data about which files exactly were modified. In general, both groups have a very similar

TABLE I
RQ1: OUTCOMES OF AGENT AND HUMAN PRS

Metric	Agent	Human	p_{Holm}	Effect
Merge rate (%)	59.5	85.1	< 0.001	$V = 0.28$
Revised PRs (%)	41.3	42.9	0.17	$V = 0.02$
PRs with a human review (%)	18.1	21.1	< 0.001	$r = -0.04$
Time to merge (h, median)	2.0	4.0	< 0.001	$r = -0.09$
Commits before merge (median)	2	2	0.17	$r = 0.02$
Linked issue closed (%)	83.7	87.5	0.06	$V = 0.05$
Time to close issue (h, median)	218	402	< 0.001	$r = -0.12$

TABLE II
RQ1: LOGISTIC REGRESSION PREDICTING WHETHER A PR IS MERGED (12,102 PRS)

Factor	Odds ratio	95% CI	p
Agent PR (vs. human)	0.25	0.21–0.31	< 0.001

CI: confidence interval.

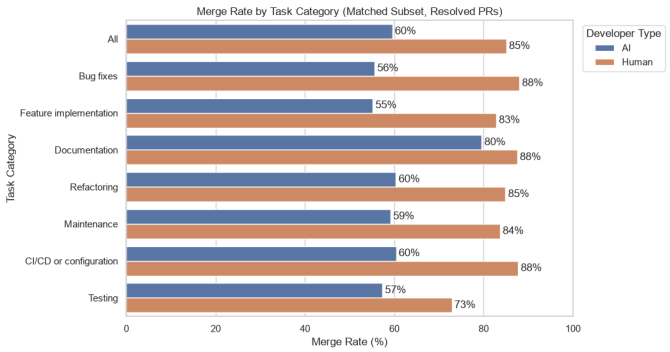


Fig. 1. RQ1: Merge rate of agent and human PRs by task category. The "All" row is the overall rate.

typical size and scope. The median number of lines added is 81 (Q1 = 17, Q3 = 361) for agent PRs compared to 76 (Q1 = 9, Q3 = 652) for human PRs ($p_{\text{Holm}} = 0.63$, $r = 0.01$), and the median code churn is 116 lines (Q1 = 26, Q3 = 537) for agents versus 123 lines (Q1 = 16, Q3 = 1,016) for humans ($p_{\text{Holm}} = 0.63$, $r = -0.01$). Agent PRs delete slightly fewer lines at the median (19 vs. 27 lines, $p_{\text{Holm}} < 0.001$, $r = -0.08$) and touch slightly fewer files per PR (median of 3 vs. 4 files, $p_{\text{Holm}} < 0.001$, $r = -0.09$), though both effect sizes are pretty negligible, and human PRs show a wider upper quartile (Q3) across all size metrics, meaning that humans tend more to sometimes create very large PRs compared to AI agents.

The bottom half of Table III and Fig. 2 show more noticeable, even though still relatively small, differences in the types of files modified and the distribution of tasks. At the file level, agent PRs touch test files a bit more often than human PRs (45.1% vs. 41.0%, $p_{\text{Holm}} < 0.001$, $V = 0.04$), but they touch documentation files less often (32.7% vs. 37.9%, $p_{\text{Holm}} < 0.001$, $V = 0.05$) and configuration files less often (31.1% vs. 42.1%, $p_{\text{Holm}} < 0.001$, $V = 0.11$). Similarly,

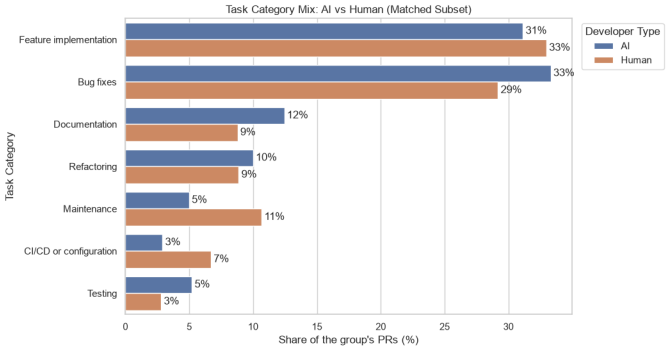


Fig. 2. RQ2: Share of each group's PRs in each task category (matched subset, all PRs).

Fig. 2 shows how all 7,541 agent and 5,565 human PRs of the matched subset are split across the seven task categories. A chi-square test across all categories shows a big difference with a small effect size ($p < 0.001$, $V = 0.16$) since feature implementation is similar (31% vs. 33%), agent PRs include a higher share of bug fixes (33.2% vs. 29.2%), documentation tasks (12.6% vs. 8.8%), and testing (5% vs. 3%), whereas human PRs include more maintenance (10.7% vs. 4.9%) and CI/CD or configuration work (6.7% vs. 2.8%).

V. DISCUSSION

a) RQ1: The biggest difference between the AI-generated and the human contributions is in pull-request acceptance, and not in what the PRs actually contain. In the matched subset, human PRs had a higher merge rate than the agent PRs (85.1% vs 59.5%), and this difference was maintained even when the task category and PR size were taken into account. Other factors, such as maintainers' trust in agent contributions or the agent's lack of project context, may therefore play a role. The same pattern is observed within the task categories, with the largest gap for bug fixes and the smallest for documentation. However, these results should not be seen as evidence that AIgenerated code is worse, since they do not take into account why a PR was not merged, which could be due to redundancy, poor timing, or other reasons unrelated to code quality.

Merged AI and human PRs required a similar number of commits, and although AI PRs were merged slightly faster, the difference was small. Accepted AI PRs also received a similar level of visible review and revision activity as human PRs. Therefore, the results do not show a substantial difference in the development process once the PR is accepted. However, reviews without inline comments and other forms of informal human oversight are not fully captured in this dataset.

b) RQ2: AI and human PRs had a similar typical size and scope. For the PRs with data at the file level, some differences were statistically significant, but their effect sizes were small. In other words, AI PRs were neither much larger or much smaller than human PRs. The difference was more

TABLE III
RQ2: SIZE, SCOPE AND FILES TOUCHED (MATCHED SUBSET, PRS WITH FILE-LEVEL DATA). p-VALUES ARE HOLM-ADJUSTED WITHIN EACH BLOCK.

	Agent			Human				
Metric	Q1	Median	Q3	Q1	Median	Q3	p_{Holm}	Effect
Lines added (lines)	17	81	361	9	76	652	0.63	$r = 0.01$
Lines deleted (lines)	3	19	137	3	27	270	< 0.001	$r = -0.08$
Code churn (lines)	26	116	537	16	123	1,016	0.63	$r = -0.01$
Files changed (files)	1	3	10	2	4	19	< 0.001	$r = -0.09$
Touches docs files (%)		32.7			37.9		< 0.001	$V = 0.05$
Touches test files (%)		45.1			41.0		< 0.001	$V = 0.04$
Touches config files (%)		31.1			42.1		< 0.001	$V = 0.11$

noticeable in the type of work. AI PRs included relatively more bug fixes and documentation work, while human PRs included more CI/CD, configuration, and maintenance work. AI agents and human developers tend to contribute to different types of tasks, rather than producing contributions of different sizes.

The file-level results provide further evidence that the two groups differ mainly in the type of files they modify. Compared with human PRs, AI PRs touched test files a little more often, and documentation and configuration files less often. Although the practical differences are limited, AI agents may be used more for testing and less for documentation and configuration-related work than human developers.

VI. THREATS TO VALIDITY

Our analysis has several limitations. Although the overall dataset is large, some agent-specific samples are small, such as Claude Code and Cursor in the matched subset. Issue-related metrics are also available for only a subset of PRs, which makes those results less representative. The available GitHub metadata is incomplete as human review states are not consistently available, some file-level details are missing due to large diffs, and the source data limits the number of commits recorded for some PRs. Reviews without inline comments and informal review activity may therefore be missed. Also, AI involvement cannot be identified perfectly from public repository data. A "human-only" PR means that no AI trace was observed rather than that AI assistance was not used. Task categories were automatically assigned and not manually validated, and documentation, test, and configuration changes were identified using file-name heuristics, which may introduce classification errors.

Finally, the findings may be specific to the repositories and time periods represented in the datasets. The AI and human groups come from different repositories, and the matched subset reduces but does not remove these differences. The GitHub data also does not contain private discussions, agent configurations, human supervision, or rejected drafts that were not submitted. Therefore, these results should not be generalized to all software projects. With more time and data, we could have manually verified the task categories, used more complete review and repository metadata, compared post-merge reverts, and measured the human effort required to supervise AI-generated contributions.

VII. CONCLUSION

In this project, we compared pull requests from AI coding agents and human contributors on a matched subset of repositories and periods. The main difference lies in acceptance as agent PRs are merged far less often than human PRs (59.5% vs. 85.1% of resolved PRs), and this gap remains after controlling for task type and PR size (odds ratio of 0.25). The gap appears in all seven task categories, with the largest difference for bug fixes and the smallest for documentation.

Beyond acceptance, the two groups look similar. Merged PRs have the same median number of commits and agent PRs are merged slightly faster. Review and revision activity is comparable, and size and scope are similar. The groups differ more in the type of work, as agents contribute relatively more bug fixes, documentation and testing, and less CI/CD, configuration and maintenance work.

Looking at the full dataset alone would give a more flattering picture of agents as 77% of resolved agent PRs are merged, but this falls to 60% on the matched subset, largely because of repositories with no human PRs, including a single personal repository that holds 27% of all agent PRs. These results show that acceptance, rather than the content of the PRs, is where agents and humans differ most. They do not measure code quality, since we do not know why a PR was rejected. Future work should manually validate the task categories, capture informal review activity, and measure the human effort needed to supervise agent contributions.

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